CLOUD COMPUTING ASSIGNMENT-2

SET UP MONITORING FOR A CLOUD-BASED APPLICATION USING AWS CLOUDWATCH

Configured Alerts and Dashboard Showcasing Metrics

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Domain	CLOUD COMPUTING

1. Introduction

Amazon CloudWatch is an AWS monitoring and observability service used to monitor cloud resources and applications through metrics, dashboards, alarms, and notifications. In this practical, an Amazon EC2 instance was selected as the monitored resource. A custom CloudWatch dashboard was created to display important EC2 metrics, and a CloudWatch alarm named “CloudWatchAlerts” was configured for CPU utilization.

The screenshots supplied for this practical document the complete monitoring sequence: the EC2 instance was prepared, monitoring metrics were displayed on a dashboard, the alarm was configured, the CPU condition was deliberately triggered, the alarm changed from OK to ALARM, the alarm history recorded the transition, and an Amazon SNS email notification was received.

2. Objective

To set up monitoring for a cloud-based application/resource using AWS CloudWatch, create a dashboard that showcases relevant EC2 metrics, configure an alert based on CPU utilization, deliberately test the alert condition, and verify the resulting alarm state and notification.

3. Cloud Monitoring Environment

Item	Configuration / Evidence
Cloud platform	Amazon Web Services (AWS)
Monitoring service	Amazon CloudWatch
Monitored resource	Amazon EC2 instance named CloudWatch-Monitoring
Instance ID	i-022dd1ea036971209
AWS Region	Europe (Stockholm) – eu-north-1
Dashboard	Cloud-Monitoring-Dashboard
Alarm	CloudWatchAlerts
Metric	CPUUtilization
Alarm condition	CPUUtilization > 20 for 1 datapoint within 5 minutes
Notification service	Amazon SNS

4. Monitoring Architecture

The implemented monitoring flow can be represented as follows:

EC2 Instance → CloudWatch Metrics → CloudWatch Dashboard

↓

CloudWatch Alarm → Amazon SNS → Email Notification

5. Monitoring Setup and Configuration

5.1 EC2 Instance Selected for Monitoring

The EC2 console shows the running instance named “CloudWatch-Monitoring”. The instance is in the Running state and has passed all three status checks. This instance is the cloud resource whose performance is monitored through CloudWatch.

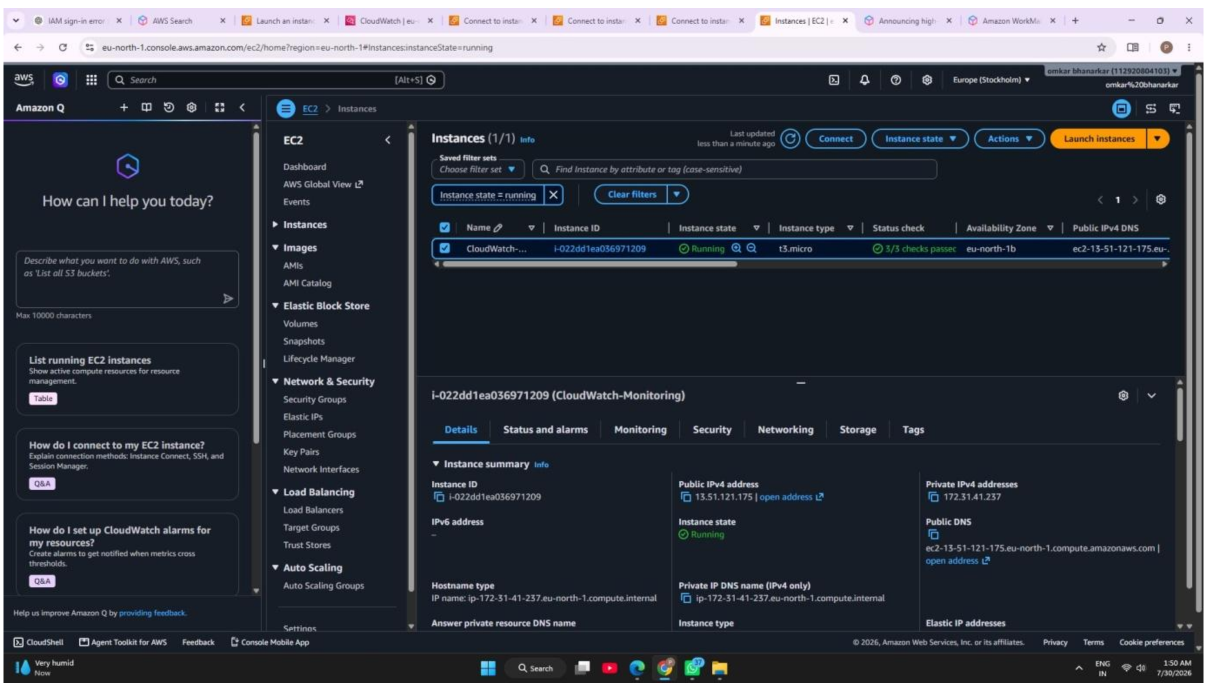

Figure 1: Running EC2 instance “CloudWatch-Monitoring” selected as the resource for monitoring.

5.2 EC2 Instance Connected and Test Environment Prepared

The EC2 Instance Connect terminal shows an Amazon Linux 2023 environment. The terminal records a stress test command using the stress utility with two CPU workers for 300 seconds. This test was used to deliberately increase CPU utilization so that the CloudWatch CPU alarm could be evaluated.

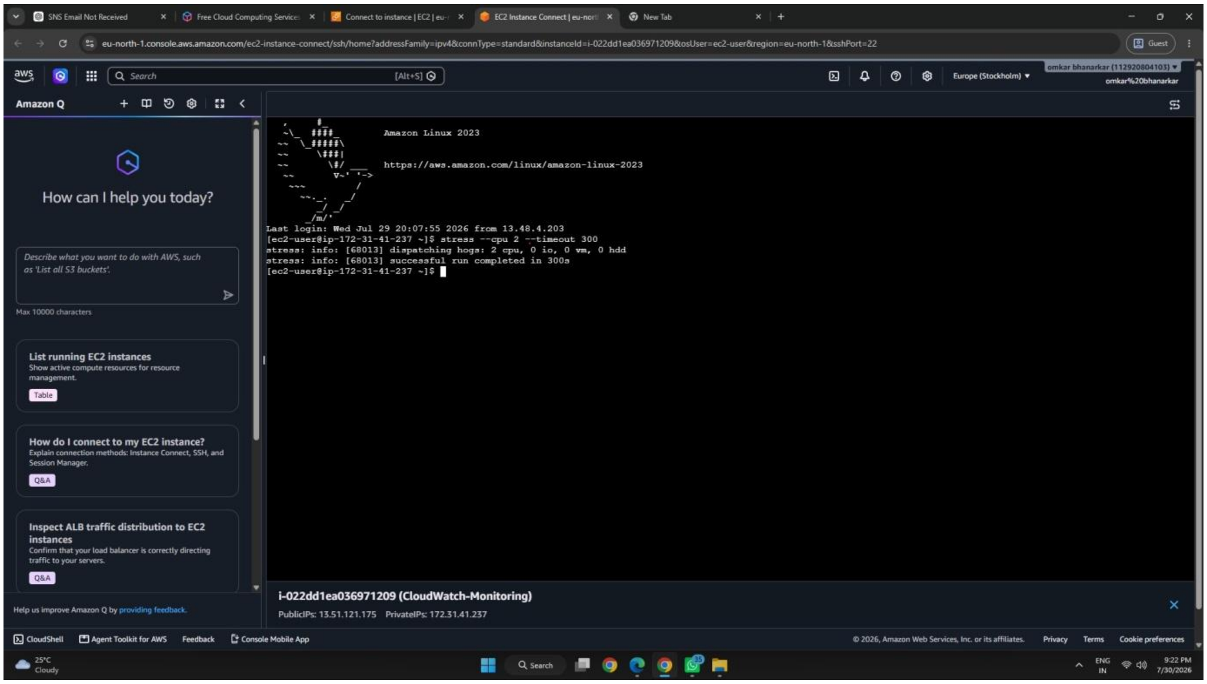

Figure 2: EC2 Instance Connect terminal showing the CPU stress test command and successful completion.

5.3 CloudWatch Dashboard Created

The CloudWatch Dashboards page shows a custom dashboard named “Cloud-Monitoring-Dashboard”. This confirms that a dedicated dashboard was created for the monitoring practical.

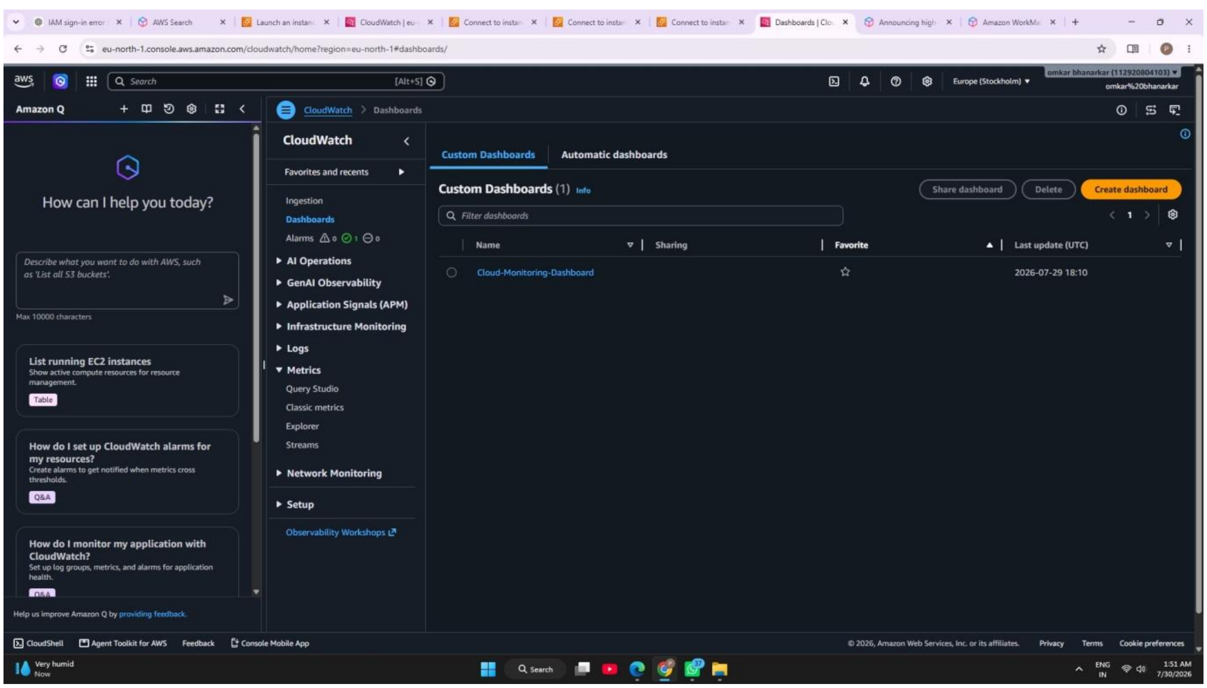

Figure 3: CloudWatch Dashboards page showing the created “Cloud-Monitoring-Dashboard”.

5.4 Dashboard Displays EC2 Metrics

The dashboard contains multiple EC2 monitoring widgets. The visible metrics include CPUUtilization, NetworkOut, NetworkIn, NetworkPacketsIn, and NetworkPacketsOut. These graphs provide a consolidated view of CPU and network activity over time.

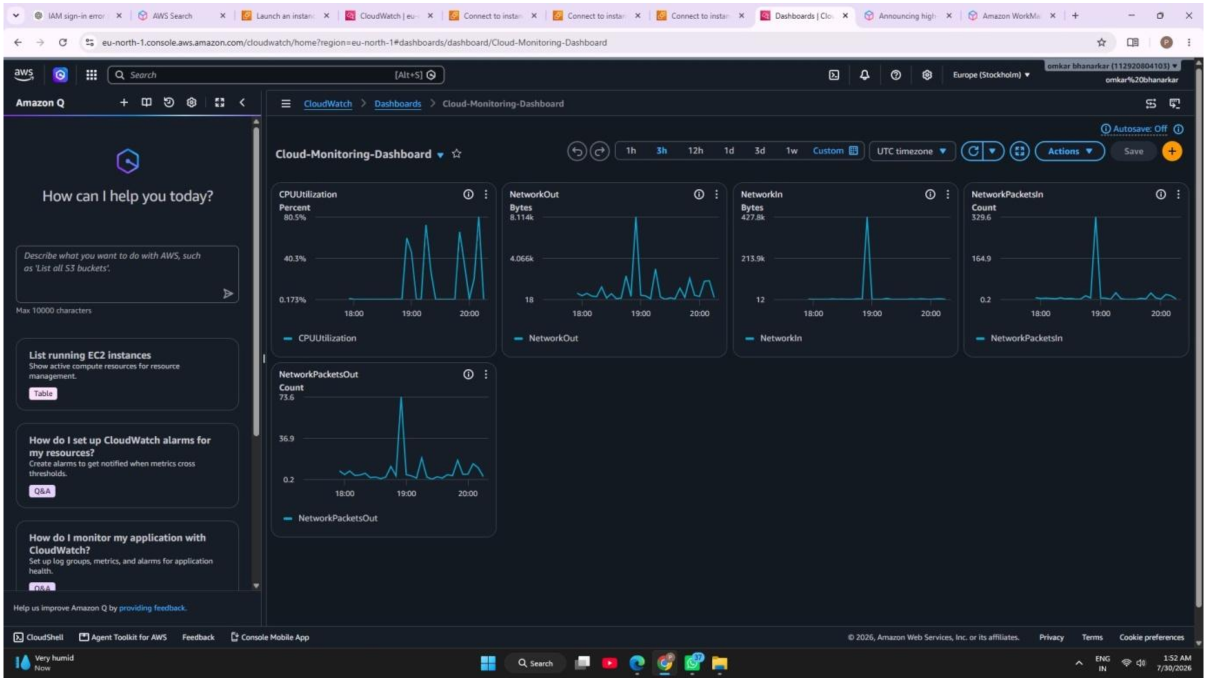

Figure 4: "Cloud-Monitoring-Dashboard" showing CPU utilization and network-related EC2 metrics.

5.5 Inspecting a Dashboard Metric

The dashboard metric can be inspected at a particular time point. In this screenshot, the NetworkOut graph is selected and a value of approximately 297.6 is displayed for the selected timestamp. This demonstrates that the dashboard can be used to examine individual metric data points.

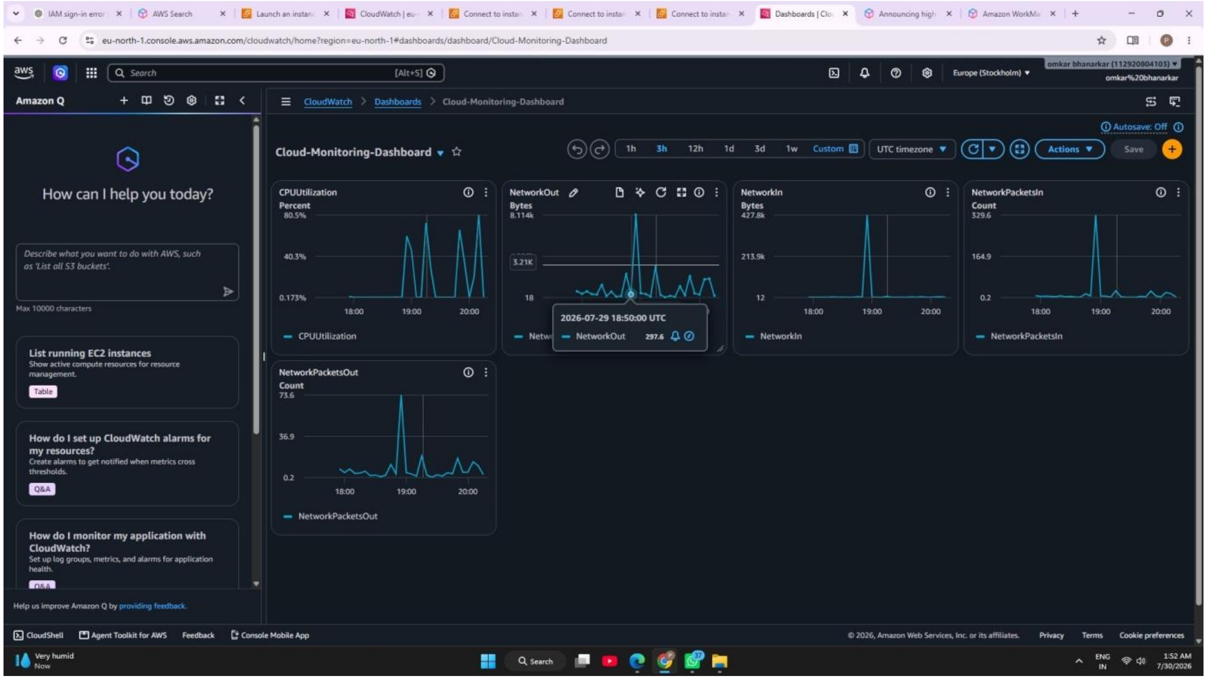

Figure 5: Dashboard metric inspection showing a selected NetworkOut data point and timestamp.

6. CloudWatch Alarm Configuration and Normal State

6.1 Alarm Created Successfully

The CloudWatch Alarms page shows an alarm named “CloudWatchAlerts”. At this stage, the alarm is in the OK state and its actions are enabled. The green confirmation banner indicates that the alarm was successfully updated.

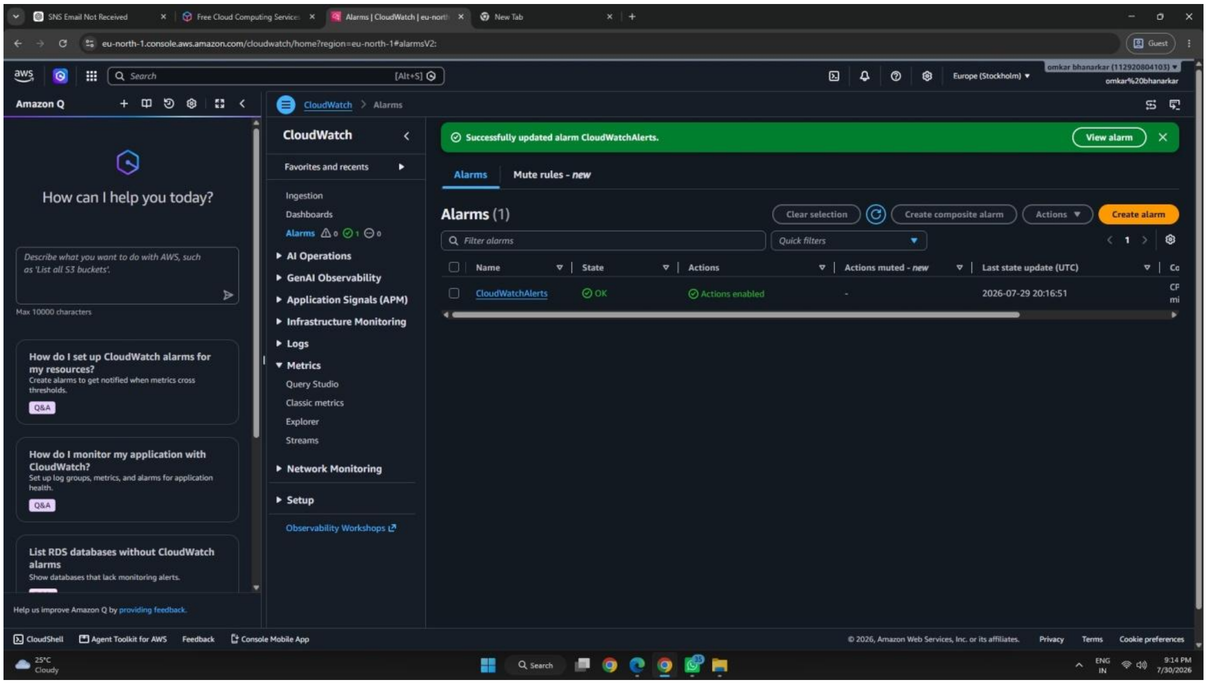

Figure 6: CloudWatch Alarms page showing the "CloudWatchAlerts" alarm in the OK state with actions enabled.

6.2 Alarm Details and Threshold

The alarm details show that the monitored namespace is AWS/EC2 and the metric is CPUUtilization for the selected EC2 instance. The alarm condition is "CPUUtilization > 20 for 1 datapoint within 5 minutes". The period is 5 minutes, the statistic is Average, and the alarm has actions enabled. This establishes the exact condition that must be crossed for the alarm to enter ALARM state.

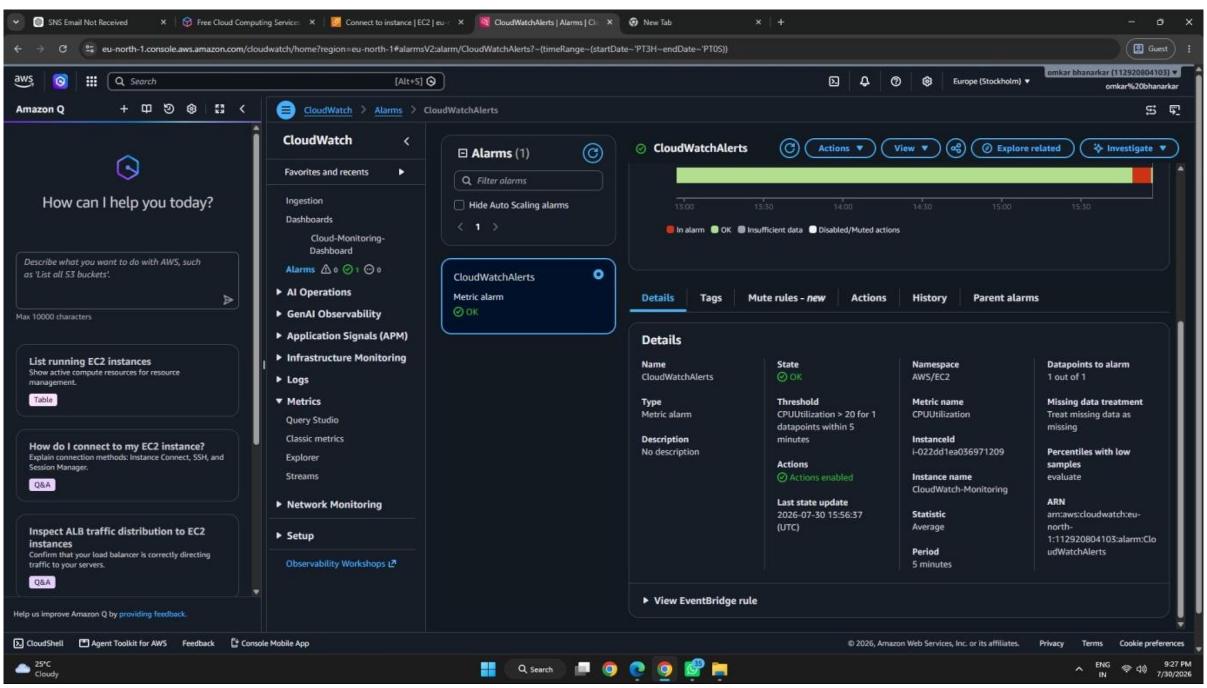

Figure 7: CloudWatch alarm details showing CPUUtilization, threshold greater than 20, 5-minute period, and enabled actions.

6.3 SNS Notification Subscription

Amazon SNS is configured as the notification mechanism for the CloudWatch alarm. The SNS topic "CloudWatchAlerts" contains one email subscription whose status is shown as Confirmed. This provides the notification destination for the alarm action.

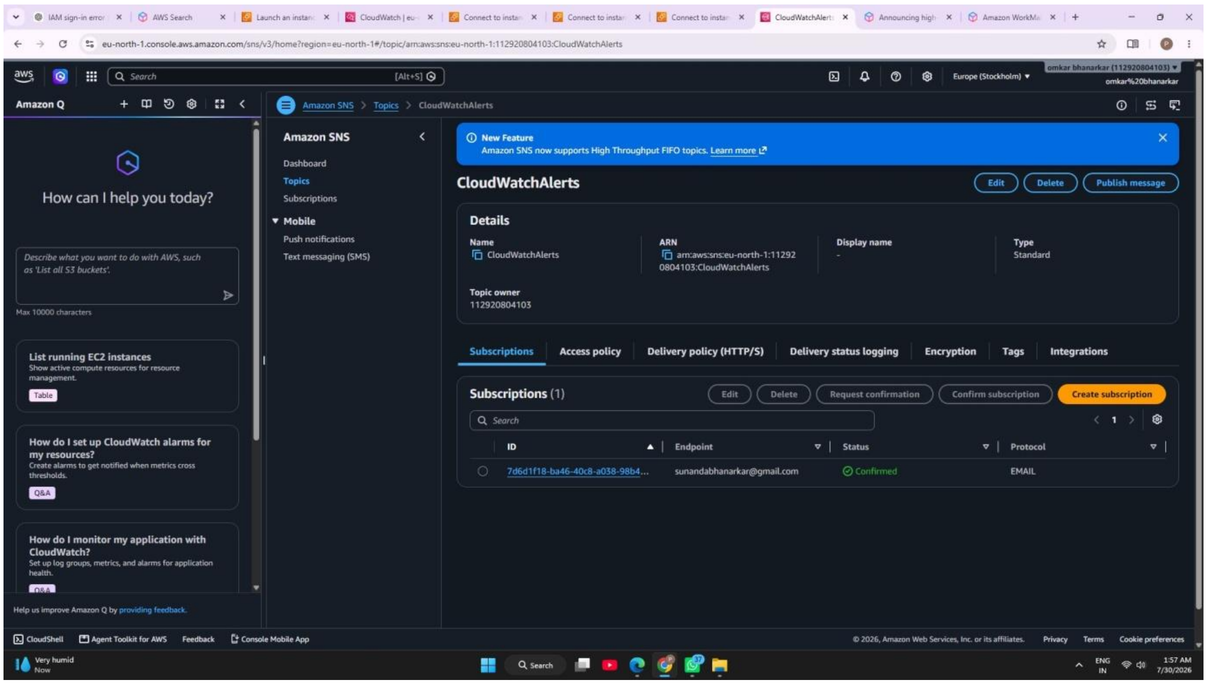

Figure 8: Amazon SNS topic "CloudWatchAlerts" showing the email subscription in Confirmed status.

7. Condition Before the Alert Was Generated

Before the alert was generated, the CloudWatch alarm was in the OK state. The alarm was configured to enter ALARM when CPUUtilization became greater than 20 for at least one datapoint within the five-minute evaluation period. The dashboard showed normal periods with low CPU utilization followed by occasional spikes.

To test the monitoring configuration, the EC2 terminal was used to run a CPU stress test for 300 seconds. The purpose of this controlled test was to raise CPU utilization above the configured threshold and verify that CloudWatch could detect the condition.

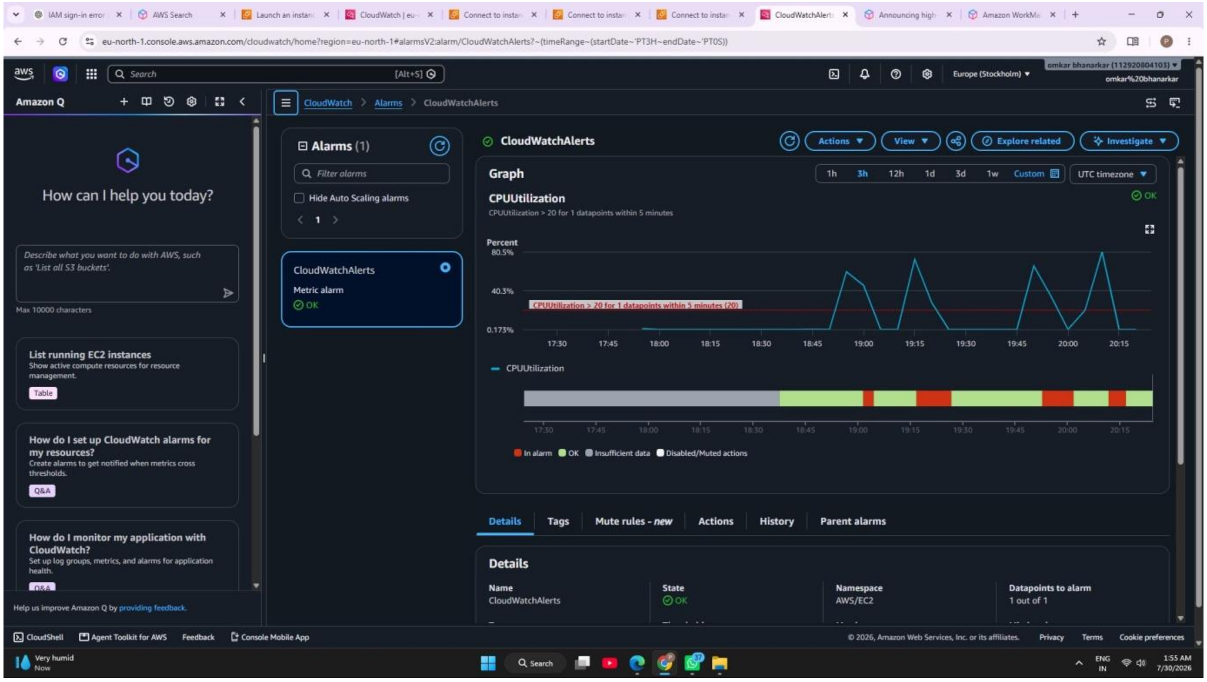

Figure 9: CloudWatch alarm in the normal OK state before the threshold-crossing event.

8. Alert Generation and State Change

After the CPU stress test increased the EC2 CPU utilization, the CPUUtilization metric crossed the configured threshold of 20. The CloudWatch alarm then changed from OK to ALARM. The alarm graph shows the CPU utilization rising above the threshold line, and the state indicator changes to ALARM.

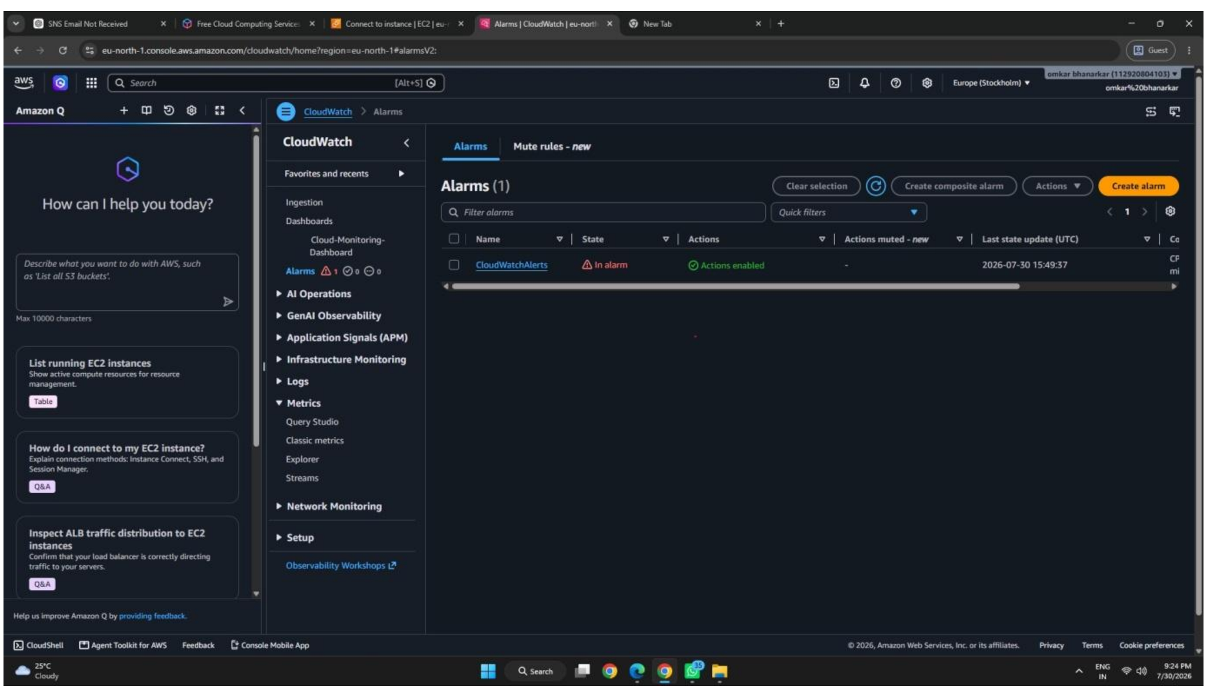

Figure 10: CloudWatch Alarms list showing "CloudWatchAlerts" in the ALARM state after the threshold was crossed.

8.1 CPU Threshold Crossed

The alarm graph provides direct evidence of the threshold crossing. The CPUUtilization graph rises sharply above the configured threshold line. The alarm state is shown as ALARM, confirming that CloudWatch detected the configured condition.

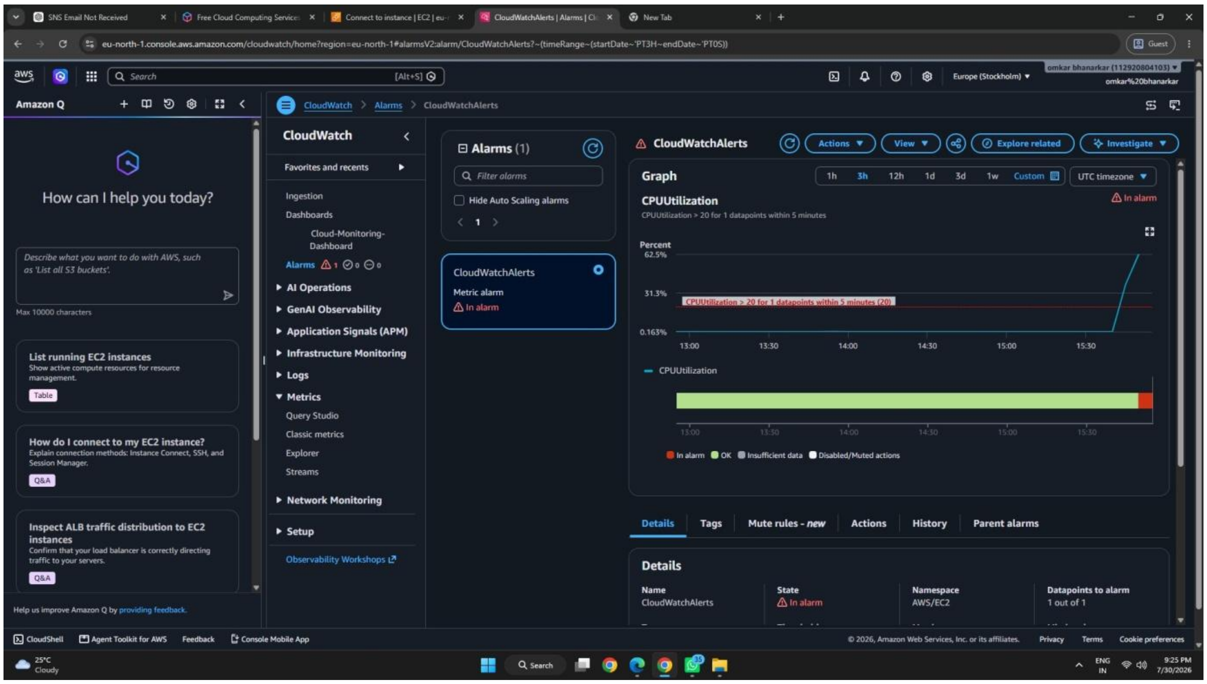

Figure 11: CloudWatch alarm graph showing CPUUtilization rising above the threshold and the alarm entering ALARM state.

8.2 Alarm Details After Triggering

The alarm details confirm the state as ALARM and show the same threshold condition: CPUUtilization greater than 20 for one datapoint within five minutes. The details identify the AWS/EC2 namespace, the CPUUtilization metric, the monitored instance, Average statistic, and five-minute period.

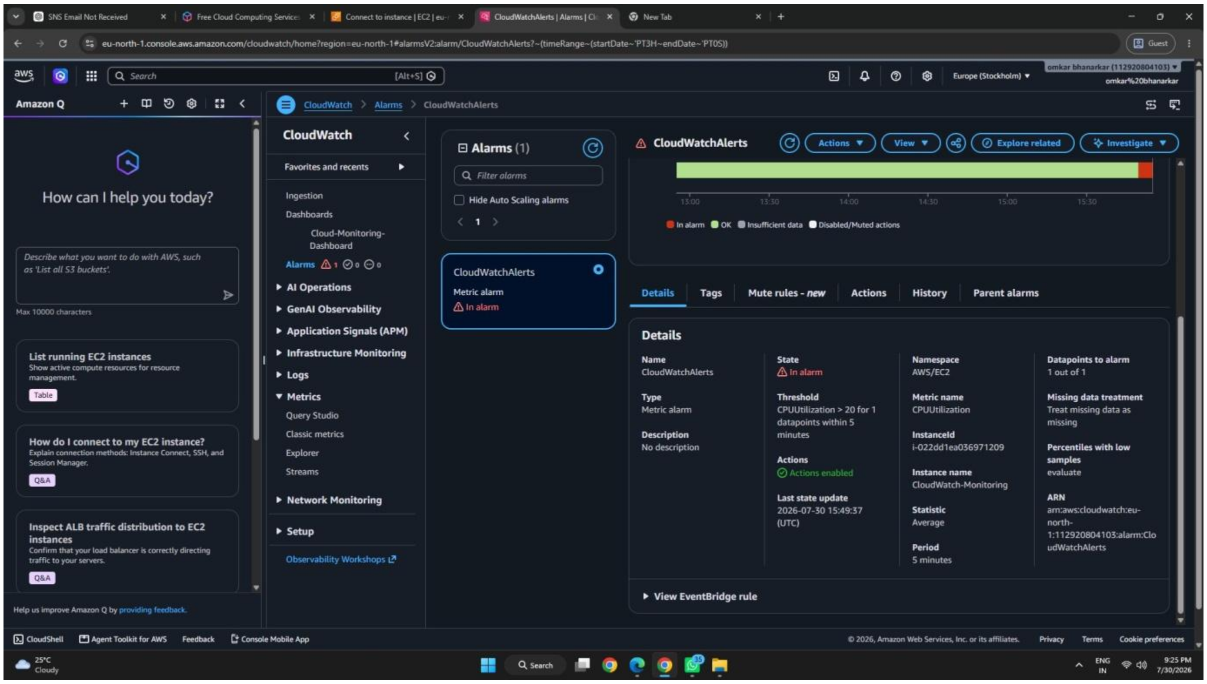

Figure 12: CloudWatch alarm details after triggering, showing ALARM state and the configured CPU threshold.

9. Alarm History and Verification

CloudWatch records state changes and alarm actions in Alarm History. The history shown in the screenshot records a state update from OK to ALARM at 15:49:37 UTC and a successfully executed alarm action. It also records earlier state changes between OK and ALARM and previous action attempts.

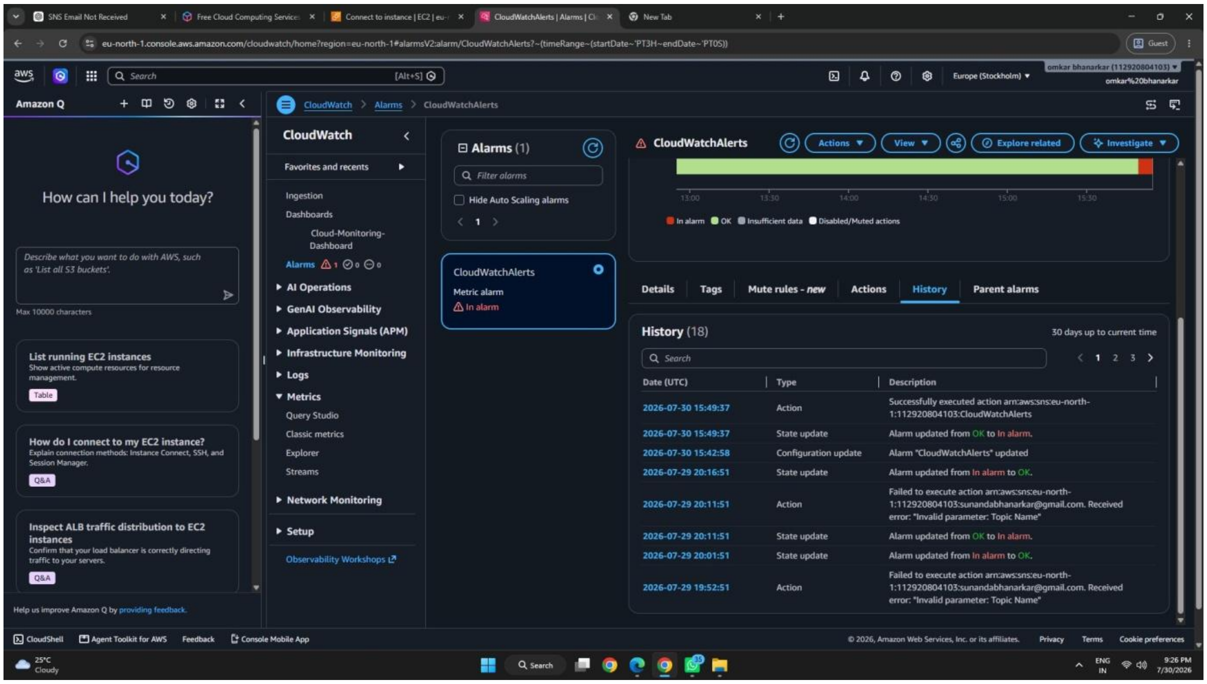

Figure 13: CloudWatch Alarm History showing the OK → ALARM transition and alarm action execution.

10. Notification Received After Alert Generation

The configured SNS email notification was received after the alarm entered ALARM state. The email states that the CloudWatch alarm "CloudWatchAlerts" entered ALARM because one of the last datapoints exceeded the threshold of 20.0. It records the state change as OK → ALARM and provides the timestamp and alarm details.

The notification confirms the end-to-end monitoring workflow: the EC2 metric crossed the threshold, CloudWatch changed the alarm state, the alarm action was invoked, and Amazon SNS delivered the alert notification by email.

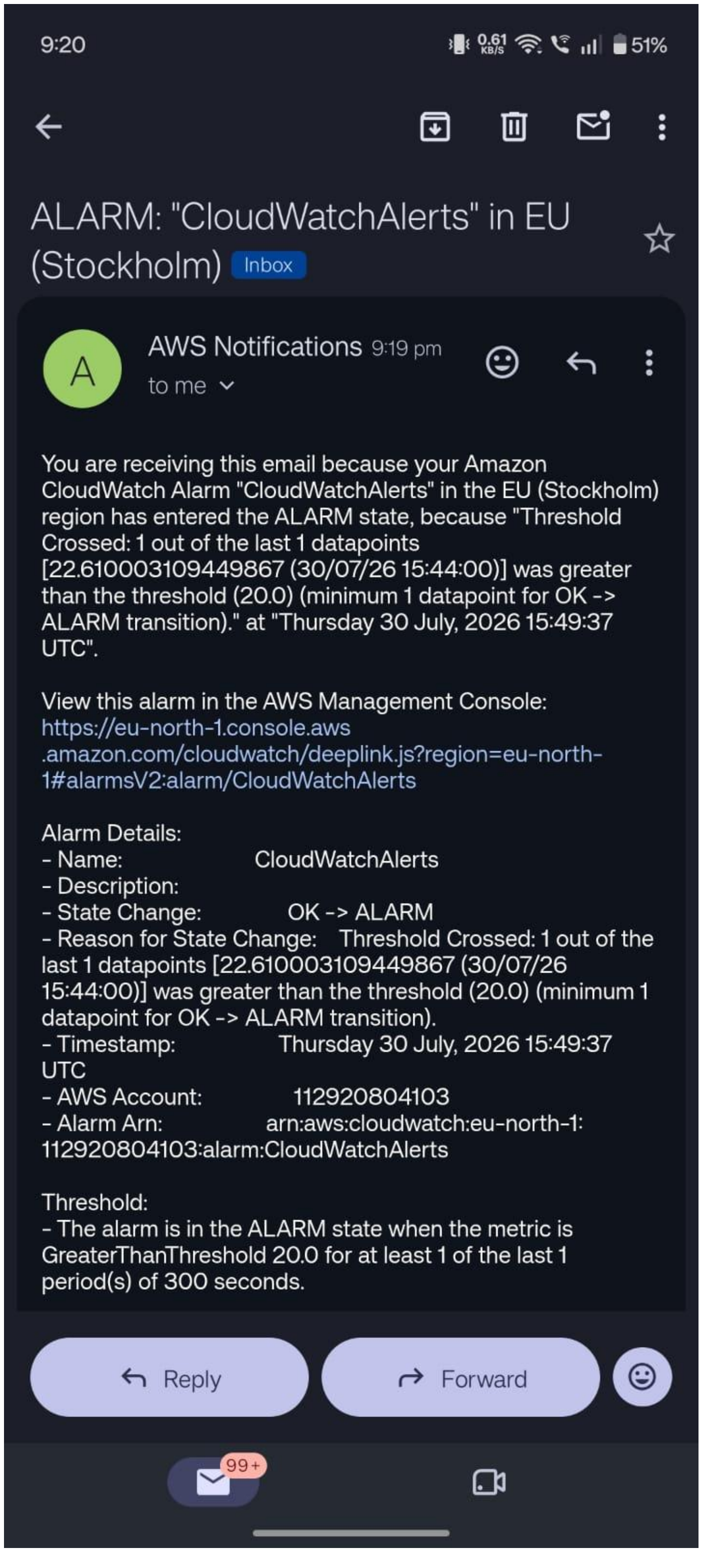

Figure 14: Amazon SNS/AWS Notifications email confirming the CloudWatchAlerts OK → ALARM state change.

11. Before and After Alert Comparison

Stage	What Happened	Evidence
Before alert	Alarm was in OK state. CPUUtilization was below the configured alarm threshold for the relevant evaluation period.	Figures 6, 7
Test action	A CPU stress test was executed on the EC2 instance for 300 seconds to increase CPU utilization.	Figure 2
Threshold crossed	CPUUtilization rose above 20, satisfying the alarm condition.	Figures 10–12
After alert	CloudWatch changed the alarm from OK to ALARM and recorded the transition in Alarm History.	Figures 11–13
Notification	Amazon SNS delivered an email explaining that the threshold was crossed and the alarm entered ALARM.	Figure 14

12. Metrics Showcased on the Dashboard

The custom CloudWatch dashboard contains several EC2 metrics. These metrics provide different views of resource activity and can help in monitoring application or instance behavior.

Metric	Purpose in the dashboard
CPUUtilization	Shows the percentage of CPU being utilized by the EC2 instance and is the metric used for the alarm.
NetworkOut	Shows outbound network traffic in bytes.
NetworkIn	Shows inbound network traffic in bytes.
NetworkPacketsIn	Shows the count of inbound network packets.
NetworkPacketsOut	Shows the count of outbound network packets.

13. Result

The AWS CloudWatch monitoring practical was successfully completed. The EC2 instance was monitored using CloudWatch, a custom dashboard named “Cloud-Monitoring-Dashboard” was created, and an alarm named “CloudWatchAlerts” was configured for CPUUtilization. A controlled CPU stress test caused the metric to cross the threshold, the alarm changed from OK to ALARM, the transition was recorded in CloudWatch Alarm History, and an SNS email notification was received.

14. Conclusion

This practical demonstrated an end-to-end cloud monitoring and alerting workflow using AWS CloudWatch. The dashboard provides continuous visibility into CPU and network metrics, while the alarm detects a predefined CPU utilization condition. The successful OK-to-ALARM transition and email notification demonstrate that the configured alerting mechanism can identify a threshold violation and communicate it to the configured recipient.

15. Submission Checklist

☑EC2 instance selected and monitored.

☑CloudWatch dashboard created with CPU and network metrics.

☑CloudWatch alarm configured for CPUUtilization.

☑Alarm threshold and evaluation period documented.

☑CPU stress test used to test the alarm condition.

☑Alarm changed from OK to ALARM after the threshold was crossed.

☑Alarm History recorded the state transition.

☑Amazon SNS email subscription was confirmed.

☑Alert notification was received by email.

☑All supporting screenshots are arranged according to the practical workflow.